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Polarization effects and the role of above-threshold ionization photoelectrons in high-order harmonic generation: comparison of experiment with quantum electrodynamic theory

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Abstract. Transition matrix elements derived previously (J Gao *et al* 1998 *Phys. Rev. Lett.* **81** 1833) from QED theory to describe high-order harmonic generation in intense optical fields are modified for elliptically polarized fields and compared with semiclassical theory and recent experimental studies of harmonic polarization characteristics. Calculations are in agreement with published data for Ne and Ar, including measurements of the components of the harmonic yield that are parallel or transverse to the major axis of the polarization ellipse for the driving optical field. The results confirm that preserving discrete photoelectron energies and incorporating two to three above-threshold ionization orders into the calculations account for the data. Also, the calculations demonstrate that the Volkov → Volkov state transition matrix element is capable of describing the polarization characteristics of the harmonics.

Above-threshold ionization (ATI) and high-order harmonic generation (HG) are related phenomena resulting from the interaction of an intense optical field with an atom or molecule. The HG spectrum of odd harmonics of the driving optical field produced by this strong field–particle coupling is characterized by a plateau, a region over which the harmonic intensity varies slowly [1]. Early experimental observations demonstrated that, while the rapid decline in the intensity of the first few harmonics can be described by perturbation theory [1], the plateau and subsequent cut-off of emission requires a non-perturbative approach. Several theoretical models have been advanced [2–5], all of which treat the optical field classically and describe the general features of the harmonics spectra. It was quickly recognized that the polarization properties of high harmonics should be relatively insensitive to propagation and phase-matching effects [6] and, therefore, offer a valuable diagnostic for studying the response of a single atom to an intense optical field [6–8]. In an effort to distinguish among the available models and, in particular, variations of the semiclassical two-step model [3, 4], a number of groups have examined in detail the polarization characteristics of harmonics in the plateau and the region above cut-off [7–11]. Measurements have included the dependence of harmonic yield on ellipticity and the angular offset between the major axes of the polarization ellipses for the harmonic and fundamental optical fields. It has become apparent that a detailed comparison of harmonic polarization characteristics with theory potentially affords, as noted in [6, 7], a tool with which to compare competing views of the HG process. Becker *et al*

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[12] have stated that '... the ellipticity and orientation of the axis of the polarization ellipse ... provide a more stringent test of any theory than just the harmonic intensities.'

Recently, we proposed [13] a QED theoretical framework for HG which predicts that harmonics are produced predominantly by a single mechanism—transitions between quantized field Volkov states which are stationary solutions of Schrödinger's equation for an electron coupled with a quantized optical field. One attractive aspect of this treatment is that the energy and momentum of all photons and electrons are clearly defined and their conservation can be followed explicitly throughout the HG process. Photoelectrons injected into the laser field by ATI have discrete energies specified by the relation

$$\frac{P_j^2}{2m_\mathrm{e}} = j\omega - E_\mathrm{b} - z\omega \tag{1}$$

where z is the ponderomotive energy U_p normalized to the fundamental photon energy $\overline{h}\omega$, E_b is the ionization potential of the atomic or molecular species, j is the number of photons absorbed and $\overline{h} \equiv c \equiv 1$. Thus, this theoretical picture of HG accounts for the experimental observation that ATI photoelectron energies are non-zero and discrete. The implications of this viewpoint when considering the harmonic polarization data in the literature are far reaching. All harmonics are generated by the electron–field interaction and the anomalies associated with particular harmonics are the result of the radiation produced by the single-atom response propagating through the background gas. In this paper, predictions of QED theory are compared with the polarization measurements of [6–9, 11, 14, 15], as well as the recent theoretical results of [15]. All of the key characteristics of experimental data are reproduced and the results confirm that, because the duration of the optical field exceeds ~100–150 fs, only the two or three lowest-energy ATI orders contribute significantly to the HG process. These calculations show that the observed polarization characteristics are introduced by the Volkov → Volkov state transition matrix element although the magnitude of the effect is weighted by the two other terms comprising the matrix for the overall HG process.

In [13], the matrix element for the generation of the qth harmonic was derived for a fundamental (driving) optical field that is linearly polarized. For a driving optical field of arbitrary polarization, the polarization vector $\hat{\varepsilon}$ is given by $\hat{\varepsilon} = \cos(\xi/2)\hat{\varepsilon}_x + \mathrm{i}\sin(\xi/2)\hat{\varepsilon}_y$, where ξ is a measure of the degree of ellipticity, μ. Defining $\mu = \tan(\xi/2)$, then $\xi = 0$ and $\pi/2$ correspond to linear and circular polarization, respectively. Therefore, equation (16) of [13] for the matrix element T_q for the overall harmonic generation process can be written as

$$\begin{aligned} T_q = \frac{eg'}{2\Lambda}\int_0^{2\pi}\mathrm{d}\varphi\int_0^{\pi}\sin\theta\,\mathrm{d}\theta\sum_j \mathrm{e}^{-\mathrm{i}j\Delta\Theta}\sqrt{2m_\mathrm{e}^3\omega^5(j-z-E_\mathrm{b}/\omega)} \\ \times (z-j)\left|\Phi_i(P_j+zk-jk)\right|^2\left|\Im_{-j}(x_0,y_0,-\phi)\right|^2 \\ \times\left\{-\frac{e\Lambda}{m_\mathrm{e}\omega}(\vec{P}\cdot\hat{\varepsilon}'^*)\Im_{-q}(x,y,-\pi/4-\phi)\right. \\ +\frac{e^2\Lambda^2}{m_\mathrm{e}\omega}(\hat{\varepsilon}\cdot\hat{\varepsilon}'^*)\mathrm{e}^{-(\mathrm{i}/2)(\Delta\Theta+\pi)}\Im_{-q+1}(x,y,-\pi/4-\phi) \\ \left.+\frac{e^2\Lambda^2}{m_\mathrm{e}\omega}(\hat{\varepsilon}^*\cdot\hat{\varepsilon}'^*)\mathrm{e}^{(\mathrm{i}/2)(\Delta\Theta+\pi)}\Im_{-q-1}(x,y,-\pi/4-\phi)\right\} \end{aligned} \tag{2}$$

where $\Im_q(x, y, \phi)$, the generalized Bessel function, is expressed as

$$\Im_q(x,y,\phi) = \sum_{m=-\infty}^{\infty} J_{-q-2m}(x)\,J_m(y)\exp(\mathrm{i}2m\phi).$$

θ and φ are the polar angles in a spherical coordinate system where θ is the angle between the momentum vector of the electron, $\vec{P}$, and $\vec{k}$, Φ_i is the initial (bound) state of the atom, Θ is the phase of the laser field, 2Λ is the amplitude of the classical vector potential, $\vec{A}(\vec{k}\cdot\vec{r})$, and g is the normalization factor arising from quantization of the field. Also, $x_0 = |(2e\Lambda/m_e\omega)\vec{P}_j\cdot\hat{\varepsilon}|$, $y_0 = -\frac{1}{2}z\cos\xi$, $x = |(4e\Lambda/m_e\omega)\vec{P}_j\cdot\hat{\varepsilon}\sin(\Delta\Theta/2)|$, $y = -z\sin\Delta\Theta\cos\xi$, $\phi = \arg(\vec{P}_j\cdot\hat{\varepsilon})$, and primes denote quantities associated with harmonics.

Before comparing this result with experiment and the predictions of semiclassical theory, it is well to first consider the role of j. At first glance, it would seem appropriate to sum equation (2) over all photoelectron orders to determine T_q. However, it is important to recall [13] that the conservation of momentum in the HG process requires that, following the emission of one or more photons by Volkov $\rightarrow$ Volkov transitions, the electron recombines with the parent atom. Electrons that do *not* return to the parent carry away momentum and energy which precludes their contribution to the harmonic spectrum. When considering the participation of photoelectron orders in HG, therefore, it is essential to take into account both the laser pulse width and electron energy. All of the HG polarization experiments reported to date have involved laser pulse widths $\geqslant$140 fs, which is sufficient for all but the lowest-energy photoelectrons to escape from the parent atom during the pulse rise time. As an example, consider harmonic generation in He for a laser intensity and wavelength of 5×10^{14} W cm^{-2} and 800 nm, respectively. Fifth-order ATI photoelectrons, having energies of $\sim$7.5 eV, travel $\sim$1100 Å in 70 fs. Hence, only the lowest photoelectron orders will interact to a significant degree with the photon field and increasing the participation of more energetic electrons in HG makes it necessary to reduce the rise time of the optical field intensity. Experimental results appear to support this conclusion. Specifically, photoelectron energy spectra acquired for high optical fields have consistently shown the comparative suppression of the lowest ATI orders relative to their higher-energy counterparts [16–19]. Mohideen *et al* [18], for example, report the observation of significantly fewer cold (<5 eV) photoelectrons in He ATI experiments than expected on the basis of KFR or tunnelling theory. In other experiments [20], the number of lowest-energy photoelectrons grows with increasing optical field intensity much more slowly than do those for the higher orders and this effect is observed in precisely the range of optical field intensities in which harmonic generation becomes significant ($>10^{13}$ W cm^{-2}).

These observations are consistent with the premise based on momentum conservation that electrons participating in HG do not appear in ATI spectra and suggest that low-energy photoelectrons contribute disproportionately to harmonic generation in the experiments conducted to date. Returning to equation (2), one concludes that, owing to physical parameters (pulse duration and intensity, electron energies), summing over all j is inappropriate when simulating existing polarization experiments. The approach adopted here is, beginning with $j_{\min}$ (the minimum number of absorbed photons necessary to photoionize the atom or molecule at a given optical intensity), to incorporate successively higher values of j into the calculation of the transition matrix T_q, while comparing the results with semiclassical theory and experiment. Regardless of the particular characteristic of harmonic generation examined (spectra [13], polarization properties, etc), the calculations repeatedly demonstrate that, for 'long-pulse' ($\tau \gtrsim$ 100–150 fs) experiments, only the lowest two to three photoelectron orders contribute significantly to harmonic generation.

Experimental studies of the effect of pump field ellipticity on harmonic generation have thus far taken essentially two forms. One is to record the yield of a specific harmonic as the ellipticity of the driving field is varied, whereas the second focuses on the characteristics of the polarization ellipse associated with each harmonic. Measurements have been reported of both the ellipticity of the harmonic radiation as well as the angle between the major axis of the polarization ellipse for a given harmonic and that for the driving (fundamental) field. Let

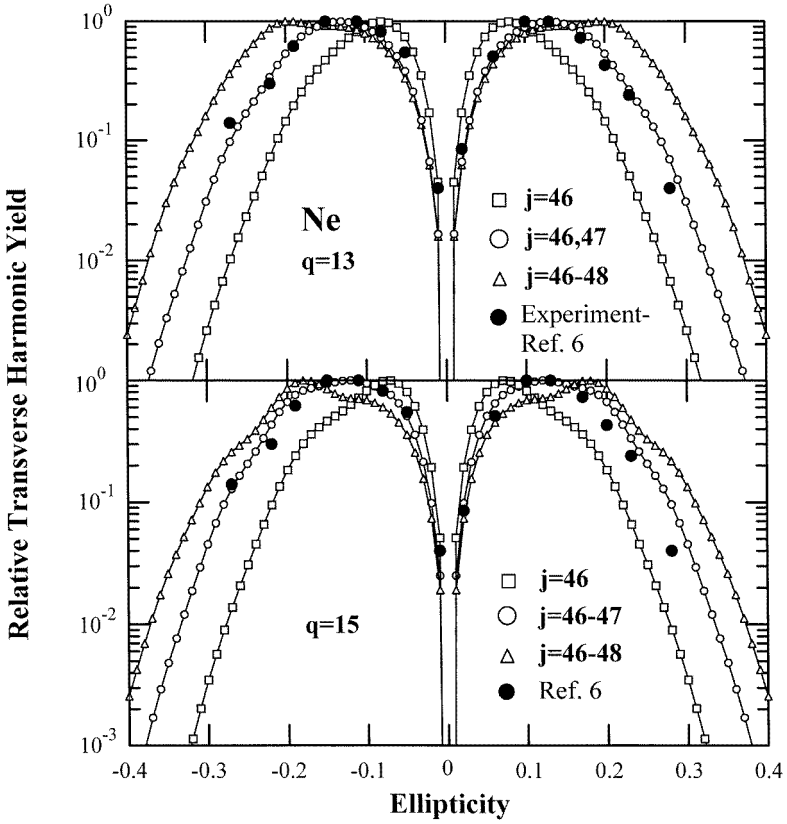


Figure 1. Comparison of the calculated and experimental (● [6]) relative yield of the transverse component of the 13th and 15th harmonics of $\lambda = 754$ nm for a driving field intensity of 10^{15} W cm^{-2}. Theoretical results are shown for three sets of values of the ATI photoelectron order j: □, $j = j_{\min} = 46$; ○, $j = 46, 47$; △, $j = 46$–48. For clarity, all data are normalized to one.

us first consider the yield data. Of particular interest is the behaviour of harmonics having photon energies comparable to the atomic ionization potential E_b. Detailed measurements of the polarization of harmonics generated in Ne have recently been reported in [6, 9, 11, 14, 15]. When ellipticity is introduced to the driving field, the yield of harmonics for which $q \approx E_b/\omega$ rises rapidly, reaching a maximum for $0.1 < \mu < 0.2$ [6, 9] or $\mu \sim 0.4$–0.5 [21]. A discrepancy between the data of Ivanov *et al* [9] and Kakehata and co-workers [21] exists insofar as the harmonic orders for which this effect is observed, but both report a minimum in the harmonic yield in the vicinity of zero ellipticity for two consecutive odd harmonics. If the components of the harmonic radiation that are polarized parallel and perpendicular to the major axis of the driving field ellipse are resolved, the result is dramatic [6]. Figure 1 compares our theoretical results with the data of [6] for the component of the 13th and 15th harmonics† of $\lambda = 754$ nm that is polarized transverse to the major axis of the driving field for an intensity of 10^{15} W cm^{-2}. Calculating the perpendicular component of the emission at the qth harmonic entails evaluating

† In [6], the data of figure 3 were attributed to the 13th harmonic but, upon re-calibrating the spectrograph, Ivanov *et al* [9] re-assigned the peak to the 15th harmonic.

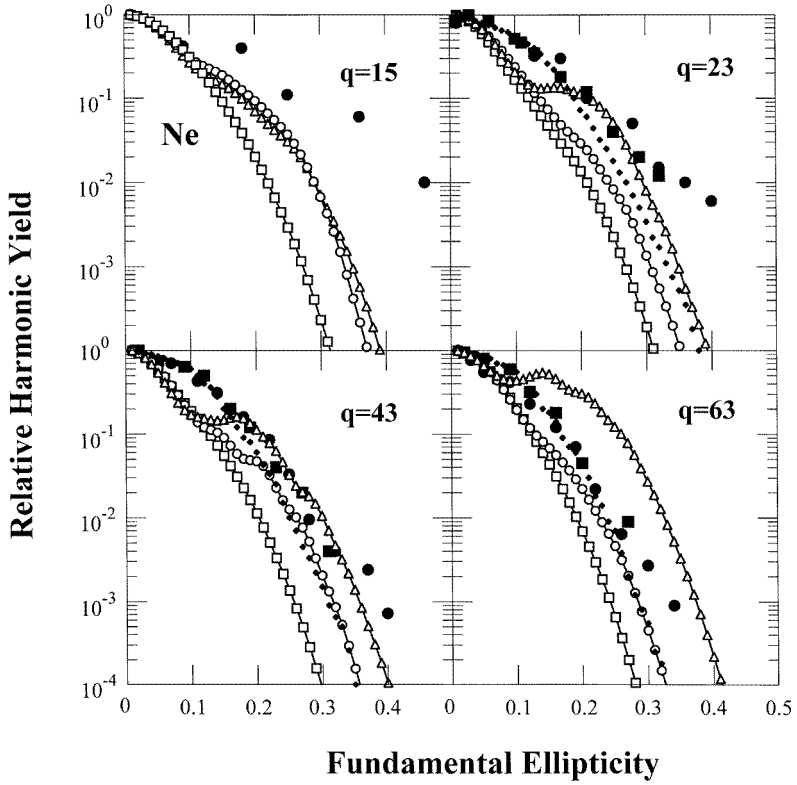


Figure 2. Dependence on ellipticity of the relative yield for the $q = 15$, 23, 43 and 63 harmonics in Ne. The fundamental optical field intensity is taken to be 10^{15} W cm^{-2}, $\lambda = 825$ nm, and the experimental results of [14] (●) and [15] (■) are shown for comparison. QED theoretical results are displayed for three sets of ATI photoelectron orders: □, $j = 57$; ○, $j = 57, 58$; △, $j = 57$–59. Also, the predictions of semiclassical theory [15], including propagation effects, for an intensity of 6×10^{14} W cm^{-2} (the largest value presented in [15]) are denoted by the full diamonds (♦).

the expression

$$Y_y^q \propto \frac{1}{2\pi} \int_0^{2\pi} \left| T_q(\hat{\varepsilon}'^* = \hat{\varepsilon}_y) \right|^2 \mathrm{d}(\Delta\Theta). \tag{3}$$

Similarly, the parallel component is given by

$$Y_x^q \propto \frac{1}{2\pi} \int_0^{2\pi} |T_q(\hat{\varepsilon}'^* = \hat{\varepsilon}_x)|^2 \mathrm{d}(\Delta\Theta). \tag{4}$$

All of the curves in figure 1 are, for convenience, normalized to one and theoretical results are shown for three sets of ATI photoelectron orders. A key result of this work is that the calculations of harmonic yield for *both* polarization components are sensitive to the degree of participation in the HG process by each ATI order. In figure 1, close agreement between QED theory and experiment occurs when the lowest two ATI orders, $j = 46$ and 47, are included, suggesting that only these lowest-energy photoelectrons contribute appreciably to the generation of $q = 13$ and 15. The same conclusion was reached in [13] regarding published harmonic spectra obtained for fundamental pulse widths >100 fs. Also, the inclusion of still higher photoelectron orders ($j > j_{\min} + 2$) results in a rapid deterioration in the fit of

calculations with experiment. Specifically, peak harmonic yield is reached at a higher value of fundamental ellipticity and the variation of harmonic response with μ broadens considerably. Note that equation (2) and the theoretical results of figure 1 account only for the single-atom response to the optical field. Although propagation effects are known to contribute significantly to harmonic spectra because of the dependence of phase matching on frequency, they play a minimal role when the variation of harmonic yield with fundamental ellipticity is under study for a specific harmonic [11, 15].

The predicted variation of the total harmonic yield with ellipticity is also in agreement with data available in the literature. Examples of the yields calculated for $\lambda = 825$ nm and $I = 10^{15}$ W cm^{-2} are presented in figure 2 and compared with the semiclassical results reported by Antoine *et al* [15] as well as the measurements of [14, 15]. The calculations shown from [15] correspond to an intensity of 6×10^{14} W cm^{-2}, the highest value for which results were given. As first reported by Budil *et al* [14], the decline of the normalized harmonic yield with increasing ellipticity becomes steeper as q rises. The $q = 15$ experimental profile is broader than that for QED theory which is likely the result of neglecting the Coulombic interaction in equation (2). This conclusion is reinforced by the pronounced narrowing observed experimentally in [14] between $q = 17$ and 19. For the higher harmonics, the shape of the calculated QED and semiclassical curves generally follows the data quite well and the semiclassical calculations agree most closely with the QED theory incorporating the two lowest photoelectron orders. Thus, the calculations and data underscore the conclusion that only

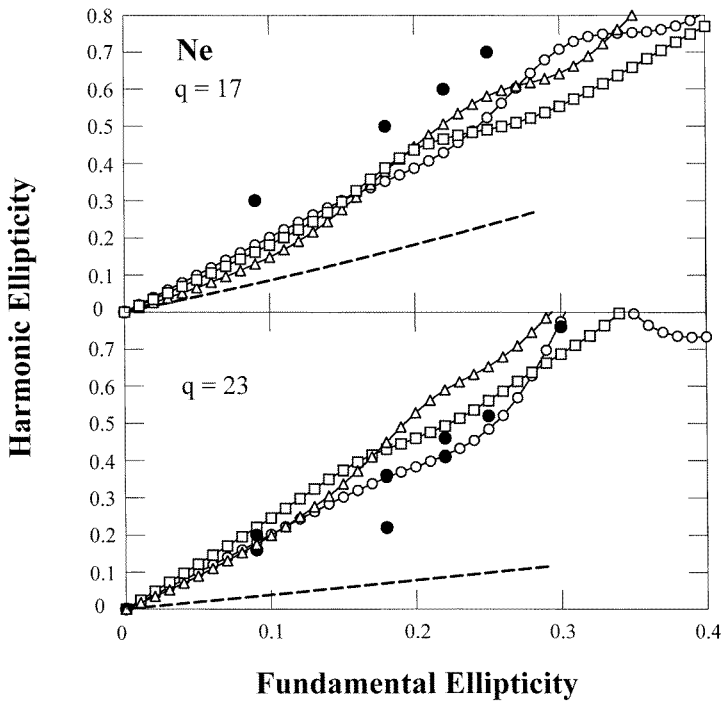


Figure 3. Calculated variation of the ellipticity of the 17th and 23rd harmonics of 790 nm in Ne with the ellipticity of the fundamental field for an intensity of 4×10^{14} W cm^{-2}. The results for the present theory (shown for three sets of photoelectron orders: ○, $j = 29$; □, $j = 29, 30$; △, $j = 29$–31) are compared with the experimental data (●) and semiclassical calculations (broken curves) of [11].

the lowest-energy ATI photoelectrons contribute significantly to HG. A similar comparison between equation (2) and the experimental results of [6, 9] for the 17th, 35th and 51st harmonics of $\lambda = 754$ nm in Ne yields the same result. Note, too, that it is the higher ATI orders (i.e. $> j_{\min} = 57$) that account for 'the departure from Gaussian at large ellipticity' that was observed in [6]. As noted earlier, the effect of incorporating increasingly energetic ATI photoelectrons into the calculations is to extend the harmonic response to larger values of ellipticity.

As a further test of the present theory, consider figure 3 which shows the predicted dependence of the ellipticity of the 17th and 23rd harmonics of 790 nm in Ne on the ellipticity of the fundamental radiation field. The calculations assumed an intensity of 4×10^{14} W cm^{-2} and invoke the relationship between the polarization vector for the harmonics and the HG transition matrix, given by the expression

$$\hat{\varepsilon}'^{*} = \frac{T_q(\hat{\varepsilon}'^{*} = \hat{\varepsilon}_x)\hat{\varepsilon}_x + T_q(\hat{\varepsilon}'^{*} = \hat{\varepsilon}_y)\hat{\varepsilon}_y}{\sqrt{T_q^2(\hat{\varepsilon}'^{*} = \hat{\varepsilon}_x) + T_q^2(\hat{\varepsilon}'^{*} = \hat{\varepsilon}_y)}}. \tag{5}$$

Results for Ne and three sets of photoelectron orders ($j_{\min} = 29 \rightarrow j_{\min} + 2$) are compared in figure 3 with the experimental data (●) and semiclassical theory (– – –) of [11]. The results, which are representative of those for other harmonics in Ar and Ne, are less sensitive than the calculations of figures 1 and 2 to the number of photoelectron orders included and reflect the experimental data quite well.

One final set of experiments with which this theory should be compared are those of Weihe and co-workers [7, 8] in which the rotation of the major axis of a given harmonic's polarization

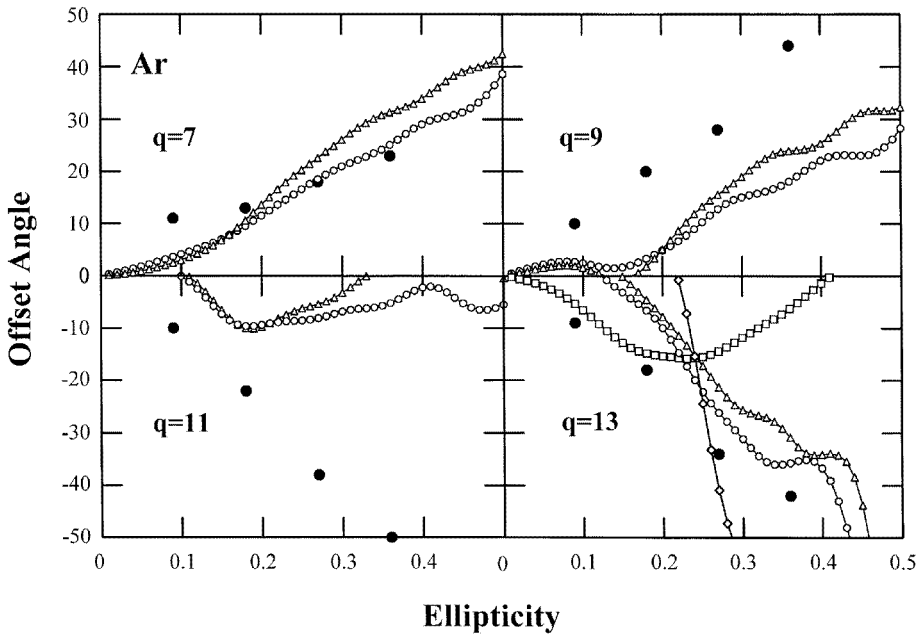


Figure 4. Offset (or rotation) angle (in degrees) between the major axis of the harmonic polarization ellipse and that for the driving optical field for four harmonics of 785 nm radiation in Ar. The results of calculations for two values of intensity (△, 4.8×10^{14} W cm^{-2}, $j = 28$–30 and ○, 5.3×10^{14} W cm^{-2}, $j = 30$–32) are presented and, for $q = 13$, the impact of incorporating one (□, $j = 29$) or four (◇, $j = 29$–32) ATI orders for an intensity of 5×10^{14} W cm^{-2} is also displayed. Data from [7, 8] are also shown (●) for comparison.

ellipse, relative to that for the fundamental, was measured as a function of ellipticity. Figure 4 compares the data of [7, 8] for four harmonics ($q = 7$–13) of 785 nm radiation in Ar with calculations incorporating as many as four ATI orders. Since the peak laser intensity for the experiments was not given, we assume I to be in the range of (1–6) $\times 10^{14}$ W cm^{-2}, based on the laser parameters and focusing geometry. When two or three ATI orders are included in the calculations (cf figure 4: $j = 28$–30 for 4.8×10^{14} W cm^{-2}, $j = 30$–32 for 5.3×10^{14} W cm^{-2}), agreement with the experimental results is obtained with the exception of $q = 11$ which is the lowest harmonic to have a photon energy exceeding the ionization potential of Ar. For all of the experimental conditions studied thus far, the calculated response for $q = 7$ and 13 best fit the data, although the theoretical results for $q = 9$ are also generally within a factor of two of experimental values. Perhaps the most significant aspect of these results is that QED theory reproduces the reversal in the sign of the offset angle between $q = 9$ and 11 that was first reported by Weihe *et al* [7, 8] in 1995. It is interesting that the difference in the polarity of the offset angle occurs for intensities above $\sim 3 \times 10^{14}$ W cm^{-2}. Below this threshold, the offset angles for $q = 9$ and 11 have the same sign. Our calculations are also consistent with the observation of Weihe and Bucksbaum [8] that the angle between the major axes for the harmonic and pump polarization ellipses does not change appreciably with laser pulse energy when ellipticity is fixed. Varying the laser intensity over the (4.8–5.3) $\times 10^{14}$ W cm^{-2} range, for example, reveals essentially the same behaviour for all four harmonics. A significant factor in the discrepancy between experimental and theoretical results for $q = 11$ is, undoubtedly, the neglect of the Coulombic interaction. The calculations of offset angle for a given harmonic are sensitive to the phase angle of T_q and, although small in *magnitude*, the Coulombic term may alter the phase of T_q appreciably. For $q = 9$ and 11, propagation effects may also play a role but the fact that the calculations reproduce the experimental trends suggests that QED theory captures the primary physical mechanisms responsible for harmonic generation.

The calculations again suggest that including only one or more than three ATI orders yields results that cannot be reconciled with experiment. As an example, the lower right-hand ($q = 13$) panel of figure 4 shows the theoretical prediction for $I = 5.0 \times 10^{14}$ W cm^{-2} if one or four ATI orders ($j = 29$–32) are included in the calculations. Note that incorporating four ATI orders leads to a reversal of the offset angle polarity for $\mu \lesssim 0.2$. Our results are also in agreement with the more recent data of the dependence of offset angle on the fundamental field ellipticity that was reported in [11] for Ne and Ar.

Perhaps the single most important conclusion of these calculations is that they show that the matrix element describing transitions between quantum Volkov states, $\langle \vec{P}', n'; 1|V'|\vec{P}, n; 0\rangle$ [13], is capable of reproducing the experimentally observed polarization characteristics of individual harmonics. Specifically, the rotation of the polarization ellipse and changes in the degree of ellipticity that occur during harmonic generation arise primarily from transitions between stationary states formed by the interaction of an electron with the photon field.

In summary, the predictions of QED theory regarding the polarization characteristics of high harmonics have been compared with experiment. The calculations generally agree with experimental measurements of harmonic ellipticity, offset angles, and parallel and transverse harmonic yields as a function of fundamental field ellipticity. To our knowledge, the data of figures 1 and 4 have not previously been reproduced theoretically. One asset of the QED approach is that the discrete photoelectron energies produced by ATI are preserved throughout the HG process and it has become clear that no more than two or three ATI orders will account for published data from experiments in which laser pulse widths exceed $\sim$100 fs. This suggests that harmonics can be generated more efficiently if a means is devised to trap higher-order ATI photoelectrons in the optical field and may account for the increased harmonic photon energies and efficiencies observed [22] as the fundamental pulse width is progressively decreased.

Acknowledgments

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